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T1A.3

Advances in Electrical Grid Assets Inspection:

Exploring Multimodal Large Language Models

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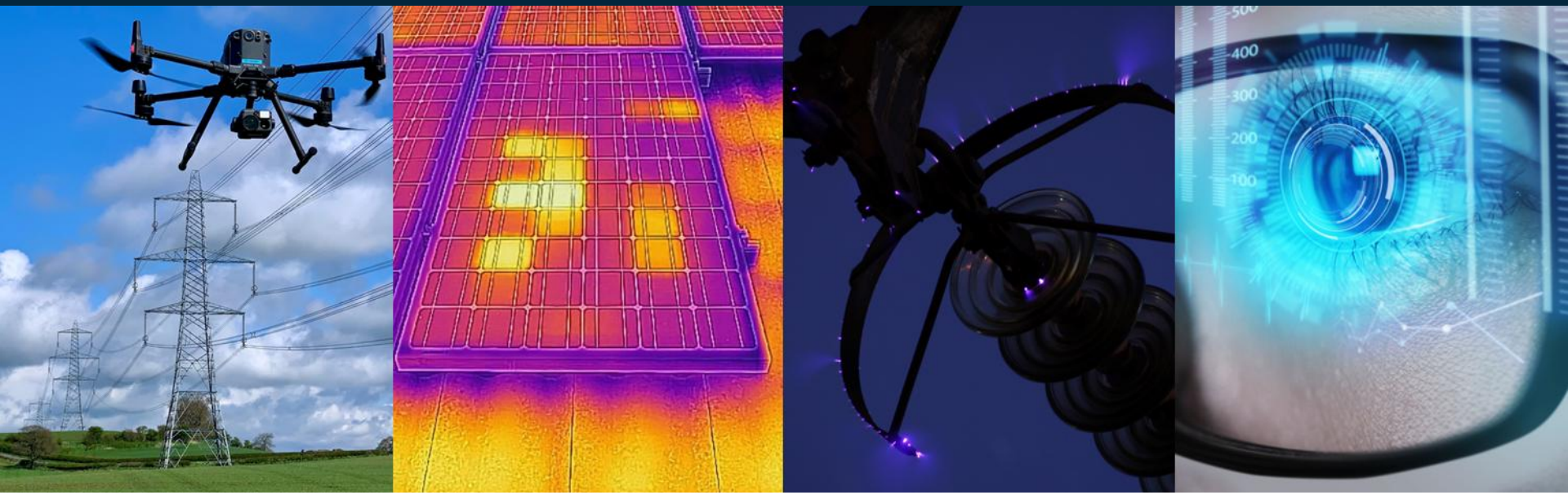
INTRODUCTION

- Motivation
- Context
- Objectives
- Overview



PROBLEM

- Electrical Power Growing Demand
- Grid Reliability and Generation Efficiency
- Maintenance Periodicity and Inspection Magnitude



SOLUTION

- Remote Visual Inspections Outsourcing
- Invisible Electromagnetic Spectrum Technologies
- Image Processing Effective Methods

Electrical Grid Asset Inspection Current Workflow



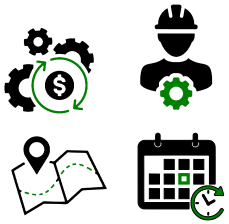
Assets Management Strategy

Power Lines
- Medium Voltage
- High Voltage
- Very High Voltage

Substations
Solar Plants
Wind Turbines
Hydroelectric Dams

Electrical Grid Asset Inspection Current Workflow





Assets Management Strategy

Inspection Work Plan

Power Lines
- Medium Voltage
- High Voltage
- Very High Voltage

Substations
Solar Plants
Wind Turbines
Hydroelectric Dams

Employed Resources
- Foot Patrol
- Helicopter
- UAV

Assets Location
Work Schedule

Electrical Grid Asset Inspection Current Workflow



Assets Management Strategy

Power Lines
- Medium Voltage
- High Voltage
- Very High Voltage

Substations
Solar Plants
Wind Turbines
Hydroelectric Dams

Inspection Work Plan

Employed Resources
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- Helicopter
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Assets Location
Work Schedule

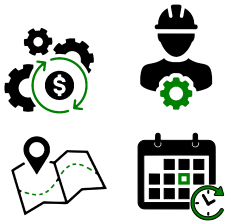
Data Collection Procedure

Technologies
- RGB
- Infra-Red
- Ultra-Violet
- LiDAR + GNSS

Data Homogeneity
Traceability

Electrical Grid Asset Inspection Current Workflow









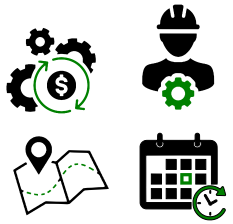
Assets Management Strategy	Inspection Work Plan	Data Collection Procedure	Data Analytic Processing
Power Lines - Medium Voltage - High Voltage - Very High Voltage Substations Solar Plants Wind Turbines Hydroelectric Dams	Employed Resources - Foot Patrol - Helicopter - UAV Assets Location Work Schedule	Technologies - RGB - Infra-Red - Ultra-Violet - LiDAR + GNSS Data Homogeneity Traceability	Defects - Missing Parts - Broken Parts - Flashed Insulators - Loose Connections - Corrosion - Dirtiness - Vegetation - Bird Nests

Electrical Grid Asset Inspection Current Workflow



Assets Management Strategy	Inspection Work Plan	Data Collection Procedure	Data Analytic Processing	Asset Quality Reporting
Power Lines - Medium Voltage - High Voltage - Very High Voltage Substations Solar Plants Wind Turbines Hydroelectric Dams	Employed Resources - Foot Patrol - Helicopter - UAV Assets Location Work Schedule	Technologies - RGB - Infra-Red - Ultra-Violet - LiDAR + GNSS Data Homogeneity Traceability	Defects - Missing Parts - Broken Parts - Flashed Insulators - Loose Connections - Corrosion - Dirtiness - Vegetation - Bird Nests	Assets - Structures - Supports - Components Severity Degree Repair Suggestion

Electrical Grid Asset Inspection Current Workflow

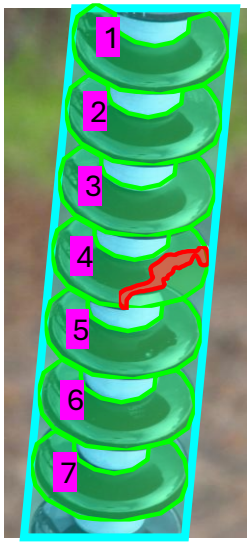




Assets Management Strategy	Inspection Work Plan	Data Collection Procedure	Data Analytic Processing	Asset Quality Reporting	Maintenance Operations
Power Lines - Medium Voltage - High Voltage - Very High Voltage Substations Solar Plants Wind Turbines Hydroelectric Dams	Employed Resources - Foot Patrol - Helicopter - UAV Assets Location Work Schedule	Technologies - RGB - Infra-Red - Ultra-Violet - LiDAR + GNSS Data Homogeneity Traceability	Defects - Missing Parts - Broken Parts - Flashed Insulators - Loose Connections - Corrosion - Dirtiness - Vegetation - Bird Nests	Assets - Structures - Supports - Components Severity Degree Repair Suggestion	Repair Replacement Reinstall Cleaning Clearing



MLLM



A vertical string of dark brown ceramic insulator discs with a glossy finish, separated by light gray cylindrical spacers. The insulator chain is part of a high-voltage power line, with a white, irregularly shaped broken edge or core exposure on one of the discs, more exactly on the fourth disc, counting top-down.

GOAL

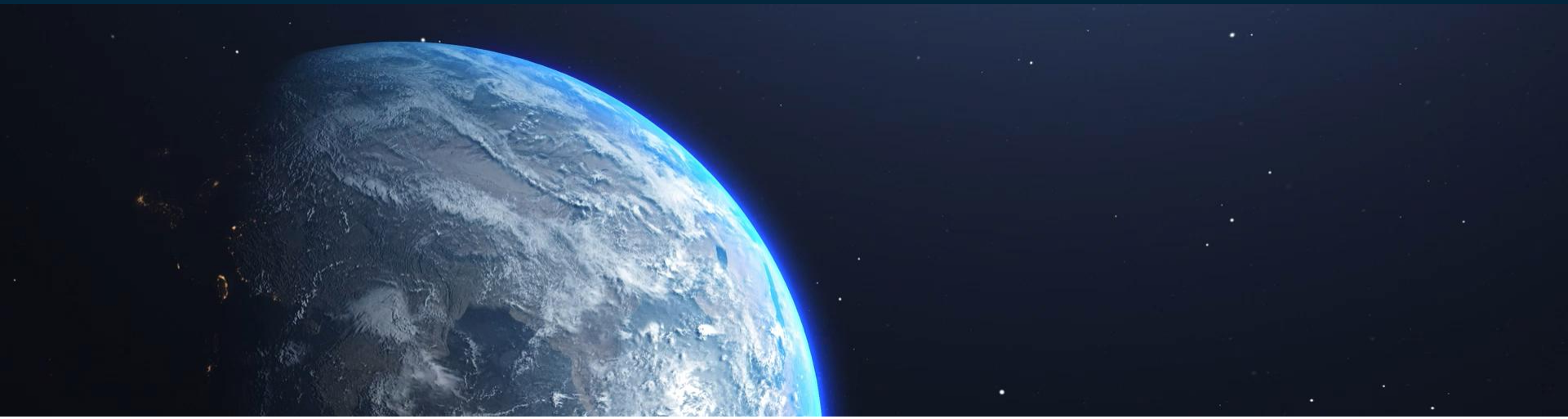
Image Processing Tool

- Assets Segmentation
- Defects Identification
- Textual Description



OBJECTIVES

- Understand Multimodal Tasks
- Identify Potential Limitations
- Fine-Tune a Multimodal LLM
- Evaluate Performance Improvement



Overview

- Deep Vision SOTA
- Experiment Design
- Results Discussion
- Summary and Future Work

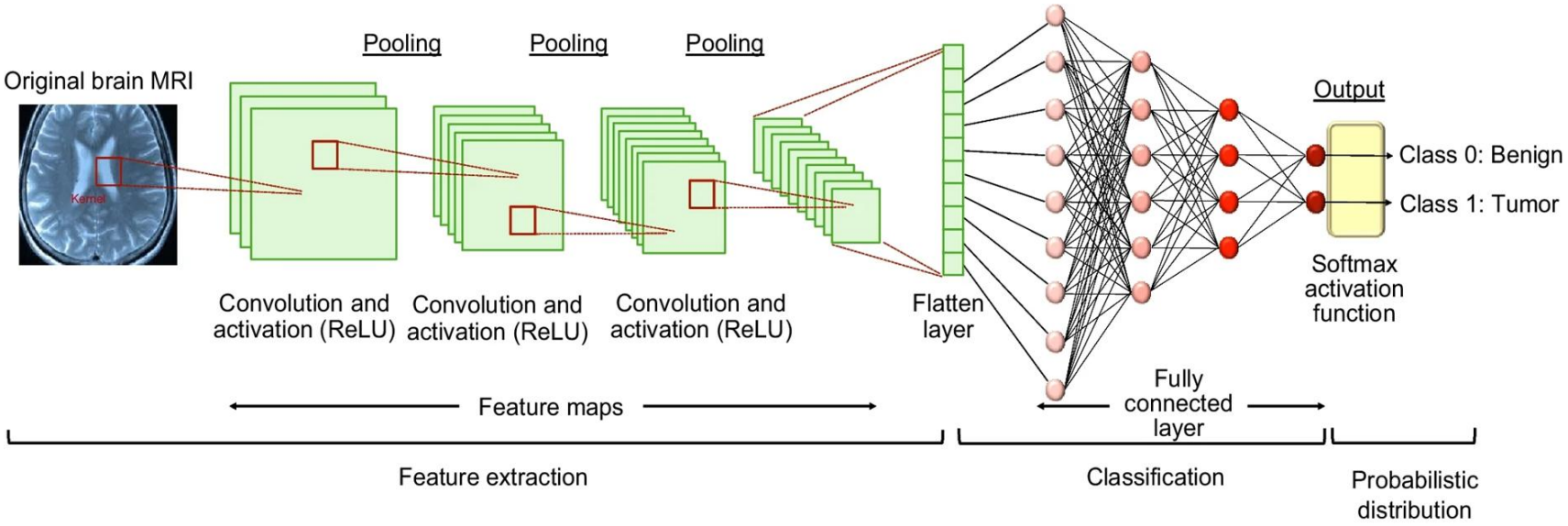
State-of-the-Art

- Deep Vision
- CNN
- ViT
- LVM & MLLM

Deep Vision Evolution

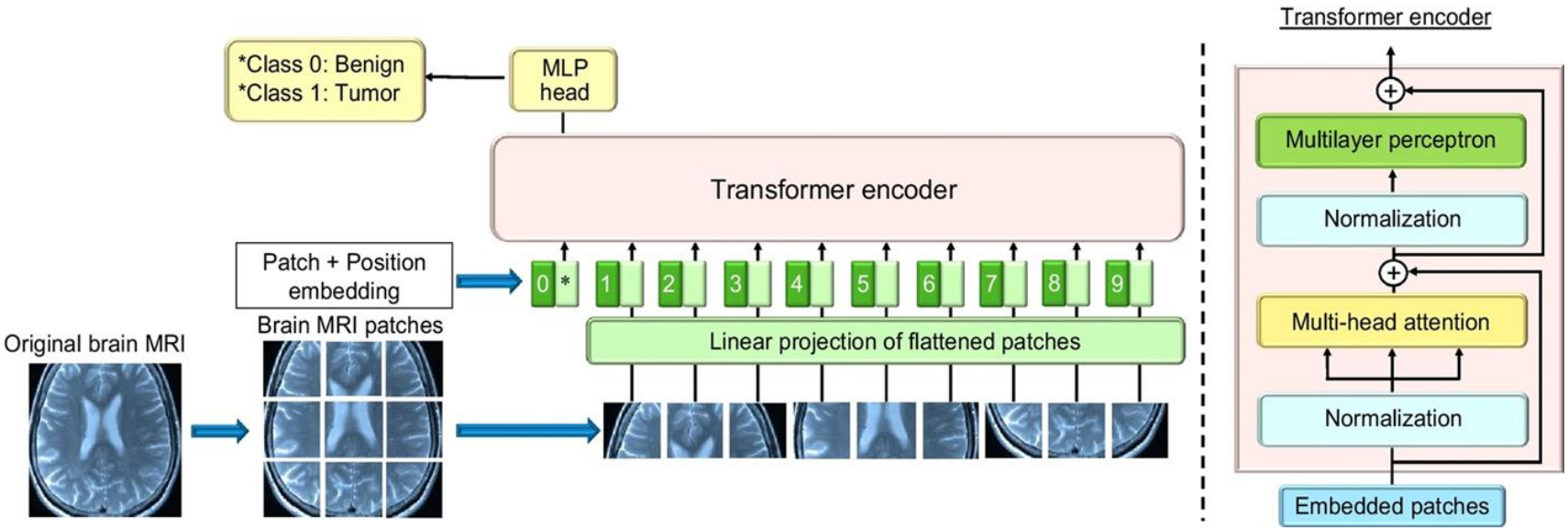


Convolutional Neural Network (CNN)



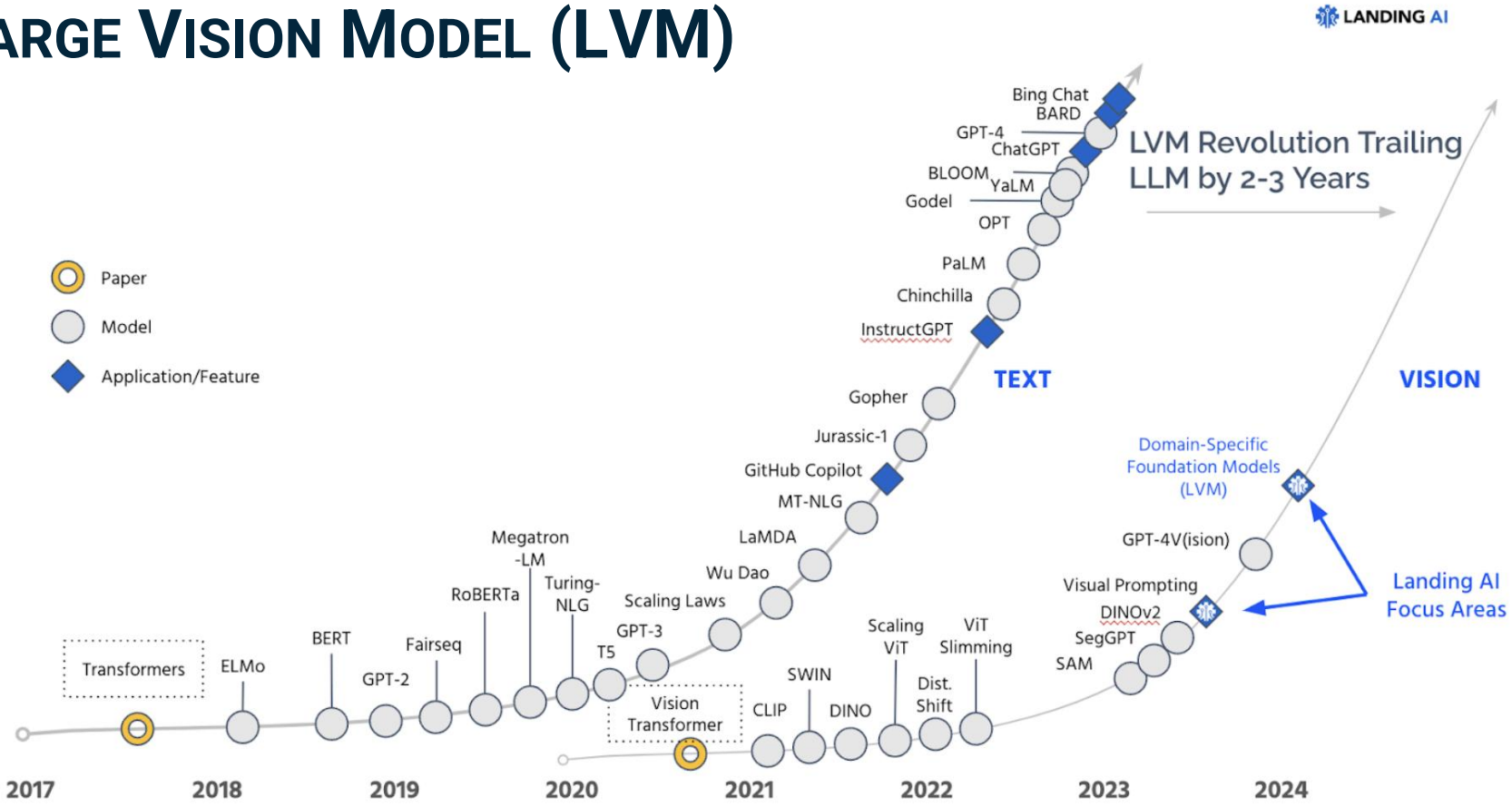
Takahashi, S., Sakaguchi, Y., Kouno, N. et al. Comparison of Vision Transformers and Convolutional Neural Networks in Medical Image Analysis: A Systematic Review. J Med Syst 48, 84 (2024). https://doi.org/10.1007/s10916-024-02105-8

Vision Transformer (ViT)



Takahashi, S., Sakaguchi, Y., Kouno, N. et al. Comparison of Vision Transformers and Convolutional Neural Networks in Medical Image Analysis: A Systematic Review. J Med Syst 48, 84 (2024). https://doi.org/10.1007/s10916-024-02105-8

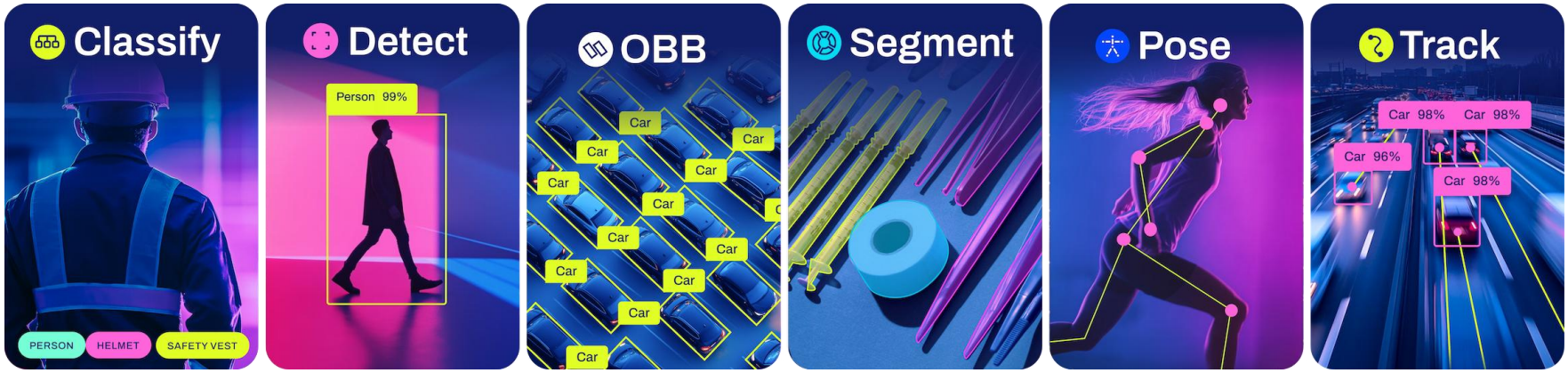
Large Vision Model (LVM)



https://landing.ai/blog/introducing-domain-specific-large-vision-models

Traditional CV Tasks

- Input X is limited to a single Image
- Target Y depends on specific the task



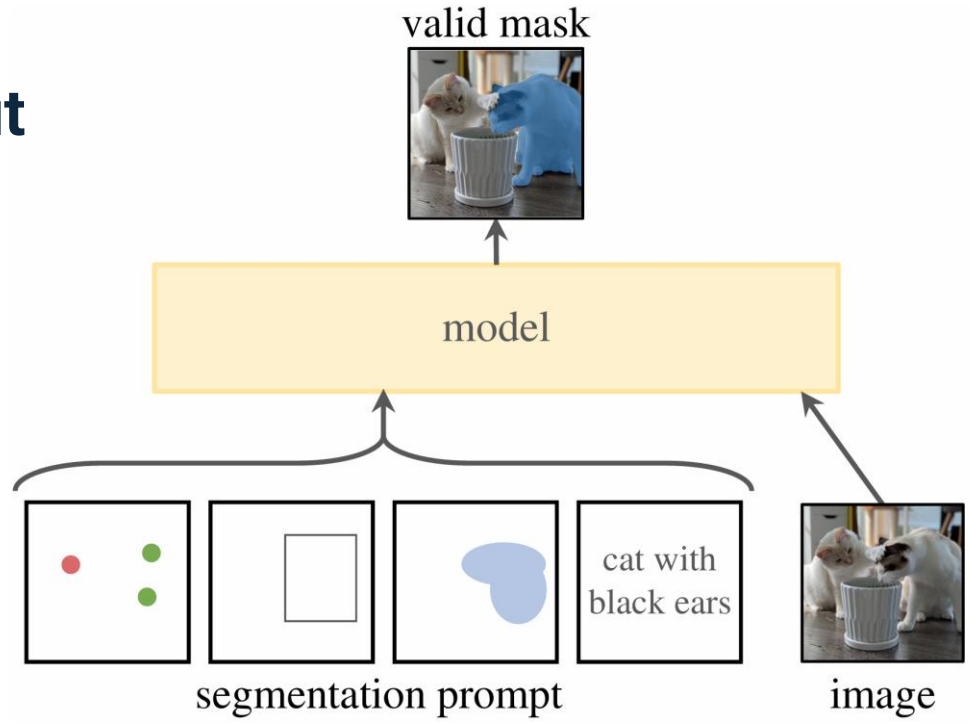
Class | Region | Oriented | Polygon | KeyPoints | Object

https://docs.ultralytics.com/tasks/

SAM: Segment Anything Model

Multimodal Prompt Input

- Positive/Negative Keypoints
- Bounding Box
- Region of Interest
- Descriptive Text

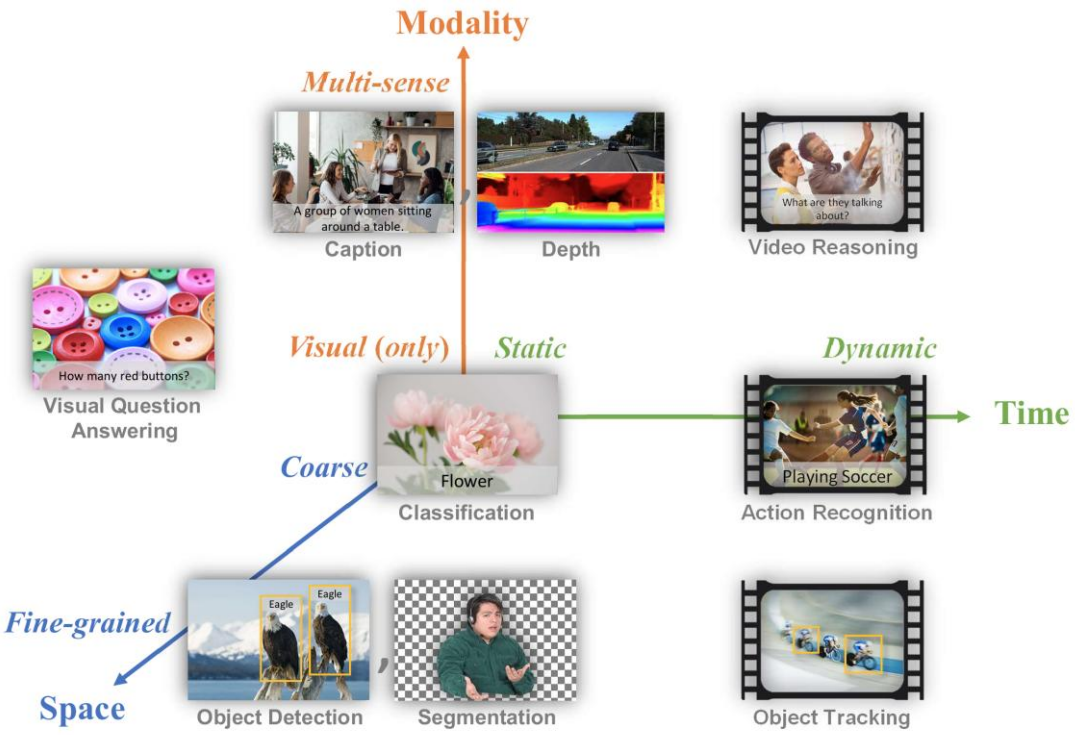


Kirillov, Alexander, et al. "Segment anything." *Proceedings of the IEEE/CVF international conference on computer vision*. 2023.

Florence: A New Foundation Model for Computer Vision

Multimodal Tasks

- Depth Estimation
- Image Captioning
- Visual Question Answering
- Video Reasoning

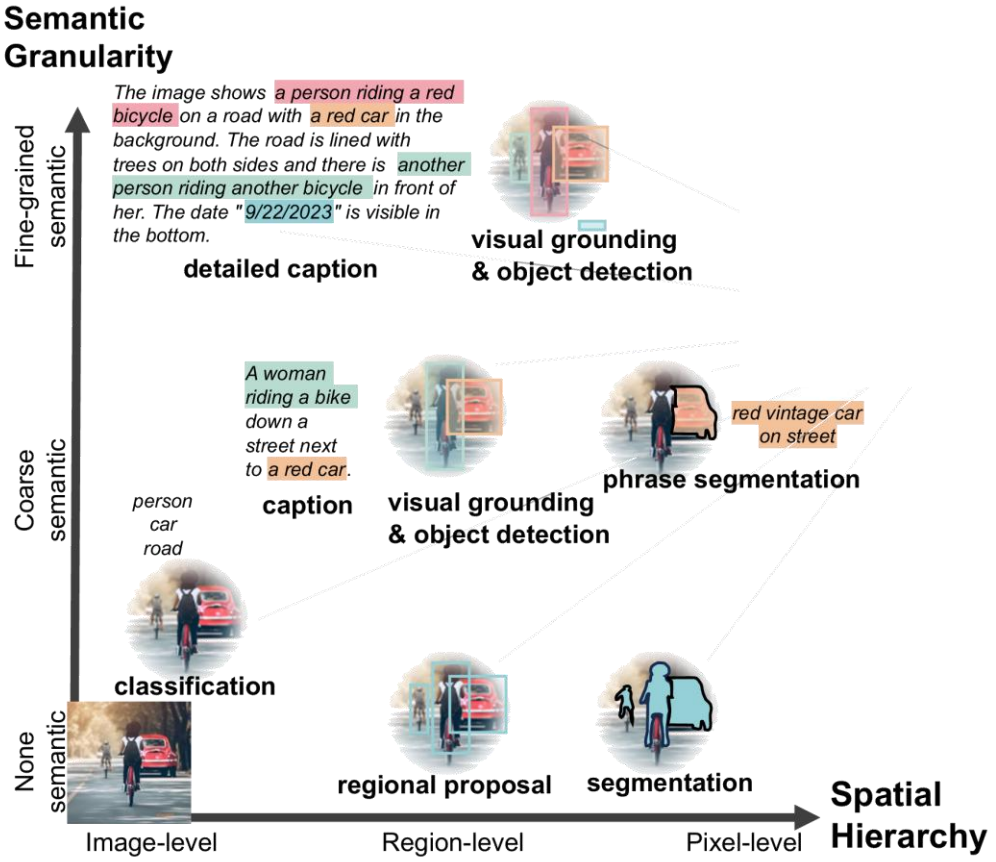


Yuan, Lu, et al. "Florence: A new foundation model for computer vision." *arXiv preprint arXiv:2111.11432* (2021).

FLORENCE-2

Multimodal Labels

- Text
- Region-Text
- Text-Phrase-Region



Xiao, Bin, et al. "Florence-2: Advancing a unified representation for a variety of vision tasks." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.

METHODOLOGY

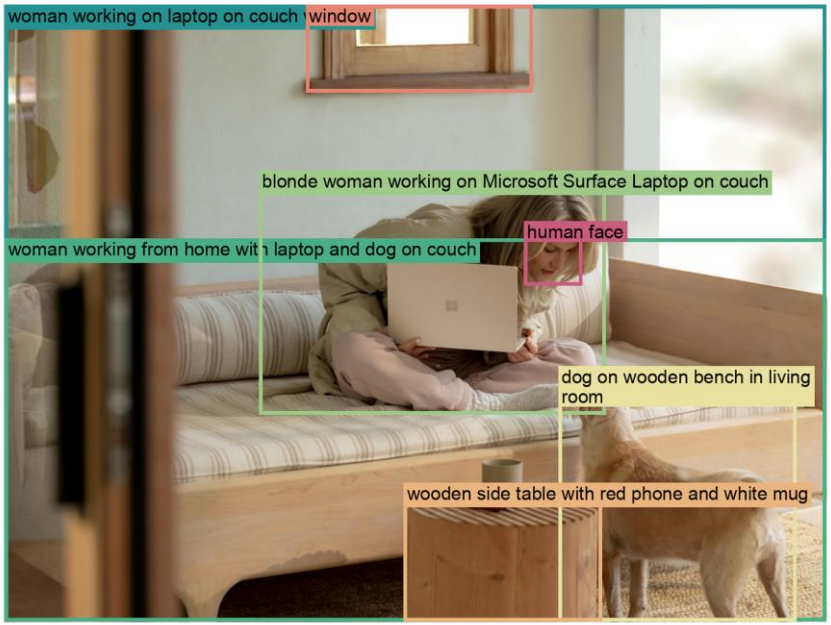
- Experimental Design
- PT and FT models
- Workstation Setup
- Dataset Description

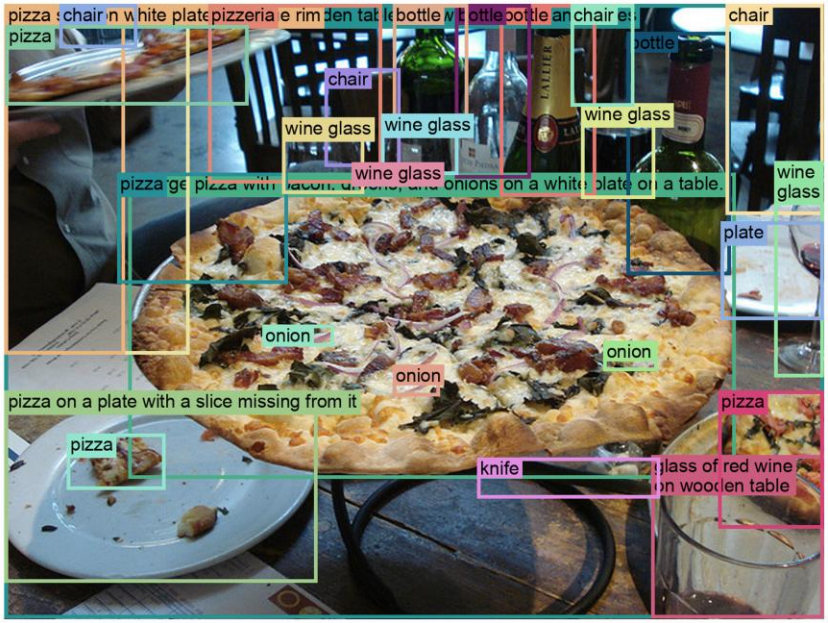
Experimental Design

	Task	Annotation Type	Prompt Input	Output
❶	Caption	Text	Image	Text
❶	Detailed caption	Text	Image	Text
❶	More detailed caption	Text	Image	Text
❷	Region proposal	Region	Image	Region
❷	Object detection	Region-Text	Image	Region, text
❷	Dense region caption	Region-Text	Image	Region, text
❸	Caption to phrase grounding	Text-Phrase-Region	Image, text	Text, region
❸	Referring expression segmentation	Region-Text	Image, text	Text, region
❸	Region to segmentation	Region-Text	Image, text	Text, region

Xiao, Bin, et al. "Florence-2: Advancing a unified representation for a variety of vision tasks." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.

	Task	Annotation Type	Prompt Input	Output
❷	Dense region caption	Region-Text	Image	Region, text

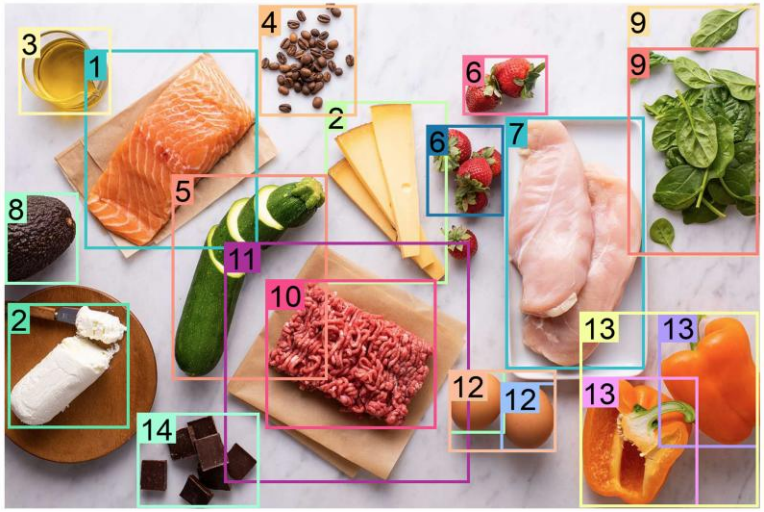




Xiao, Bin, et al. "Florence-2: Advancing a unified representation for a variety of vision tasks." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.

	Task	Annotation Type	Prompt Input	Output
❸	Caption to phrase grounding	Text-Phrase-Region	Image, text	Text, region

PROMPT: LOCATE THE PHRASES IN THE CAPTION: {CAPTION}



The image is a flat lay of various food items arranged on a white marble countertop. On the left side of the image, there is a piece of salmon[1]. Next to it, there are slices of cheese[2], a glass of oil[3], coffee beans[4], a zucchini[5], a bunch of strawberries[6], two chicken breasts[7], a avocado[8] and a few whole spinach leaves[9]. In the center of the table, there appears to be a pile of ground beef[10] on paper[11], two eggs[12], two orange bell peppers[13], and some dark chocolate bars[14]. The items are arranged in a way that suggests they are being prepared for a meal.

Xiao, Bin, et al. "Florence-2: Advancing a unified representation for a variety of vision tasks." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.

	Task	Annotation Type	Prompt Input	Output
❸	Region to segmentation	Region-Text	Image, text	Text, region





Xiao, Bin, et al. "Florence-2: Advancing a unified representation for a variety of vision tasks." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.

Pre-Trained (PT) Model

Pretrained with FLD-5B

- 126 M Images
- 500M text annotations
- 1.3B text-region annotations
- 3.6B text-phrase-region

Model	# Params	COCO Det. val2017 mAP
Florence-2-base-ft	0.23B	41.4
Florence-2-large-ft	0.77B	43.4

https://huggingface.co/microsoft/Florence-2-base-ft

Fine-Tuned (FT) Model

Finetuned with curated IDID

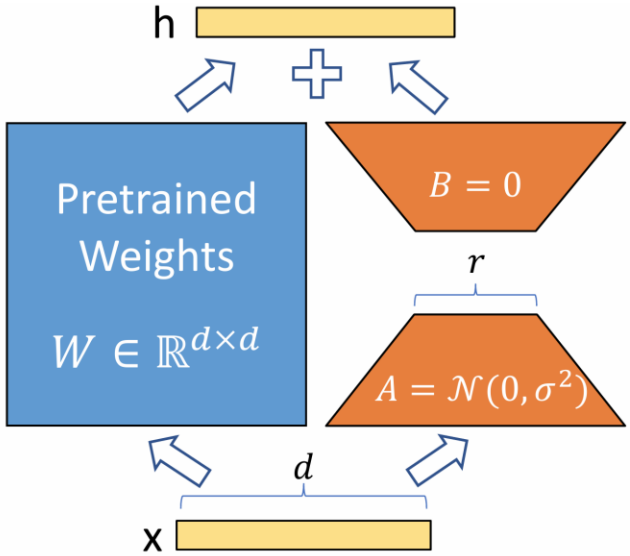
- 300 Images
- 3.1k Region annotations
- 4 Classes

Hyper-Params	Value
Epochs	100
Learning Rate	1×10^{-5}
Batch Size	8
LoRA alpha	8
LoRA rank	16
dropout	0.05

Dexter Lewis and Pratik Kulkarni, 'EPRI Insulator Defect Image Dataset'. IEEE DataPort, Aug. 11, 2021. doi: 10.21227/VKDW-X769

LoRA: Low-Rank Adaptation of LLMs

- Froze Pre-trained Weights
- Indirect Optimization of Parameters
- Merge Trainable Matrices
- Efficient Adaptation of Weights
- Lower Hardware Requirements



E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," Oct. 2021, arXiv:2106.09685. [Online]. Available: http://arxiv.org/abs/2106.09685

Workstation Setup

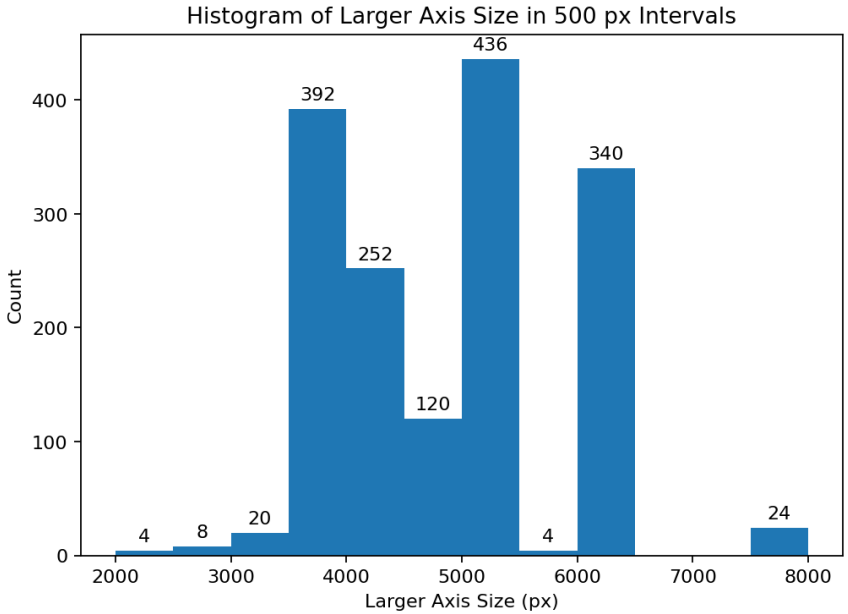
- Nvidia RTX A6000 48 Gb
- 256 Gb RAM
- 16-Core CPU
- Kubuntu 20.04
- CUDA 12.1
- Torch 2.4

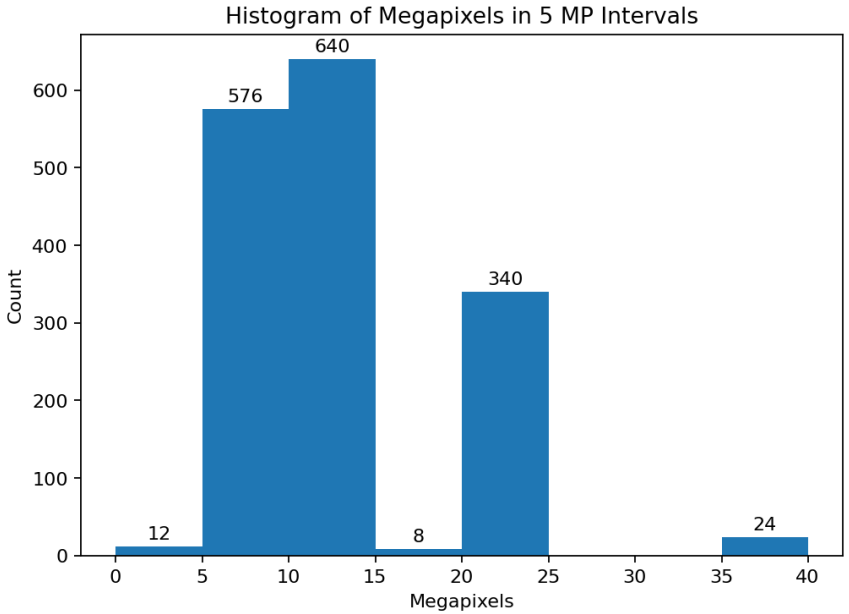


E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," Oct. 2021, arXiv:2106.09685. [Online]. Available: http://arxiv.org/abs/2106.09685

Dataset Description

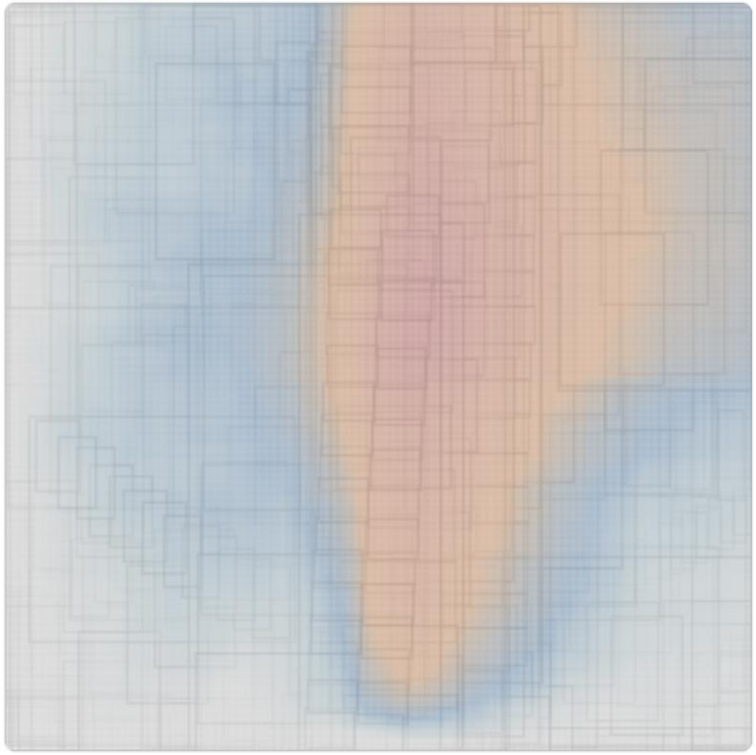
Median Image Ratio = 4416 x 3008 pixels

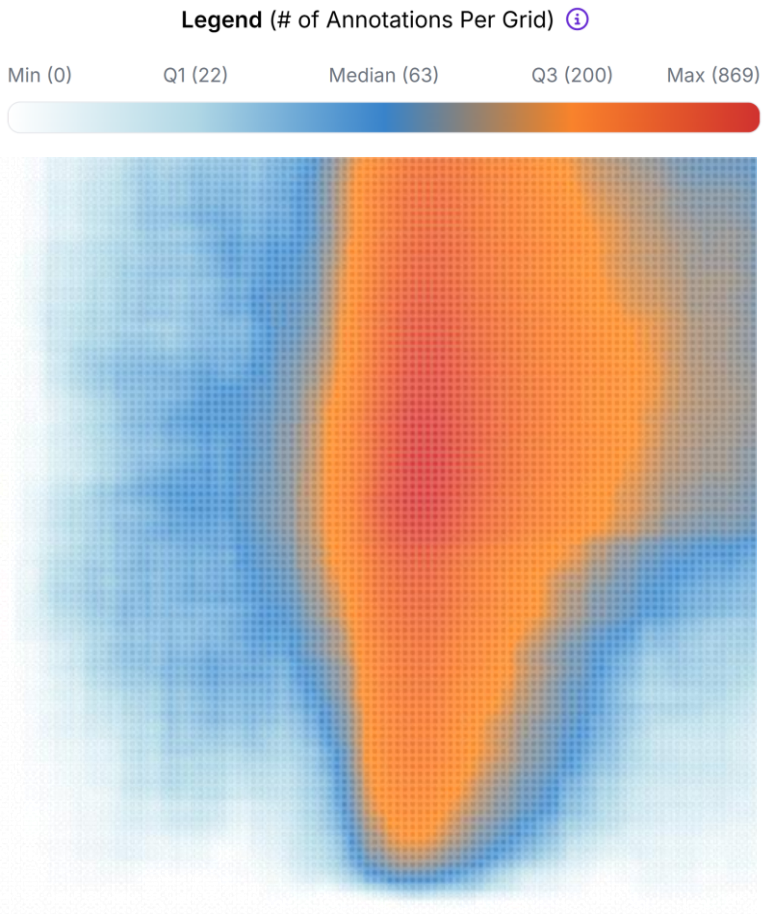




Dexter Lewis and Pratik Kulkarni, 'EPRI Insulator Defect Image Dataset'. IEEE DataPort, Aug. 11, 2021. doi: 10.21227/VKDW-X769

Annotations Heat Map

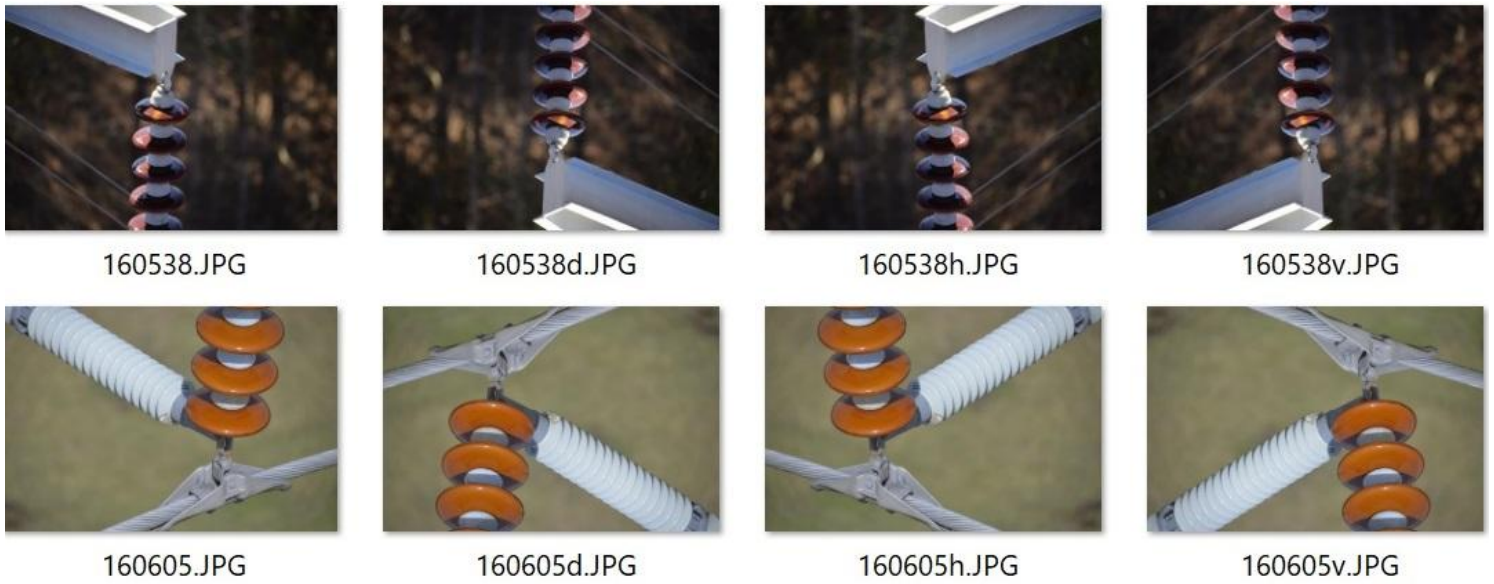




https://app.roboflow.com/academic-it/epri-insulators/health

Insulator Defect Image Dataset 1600 Images

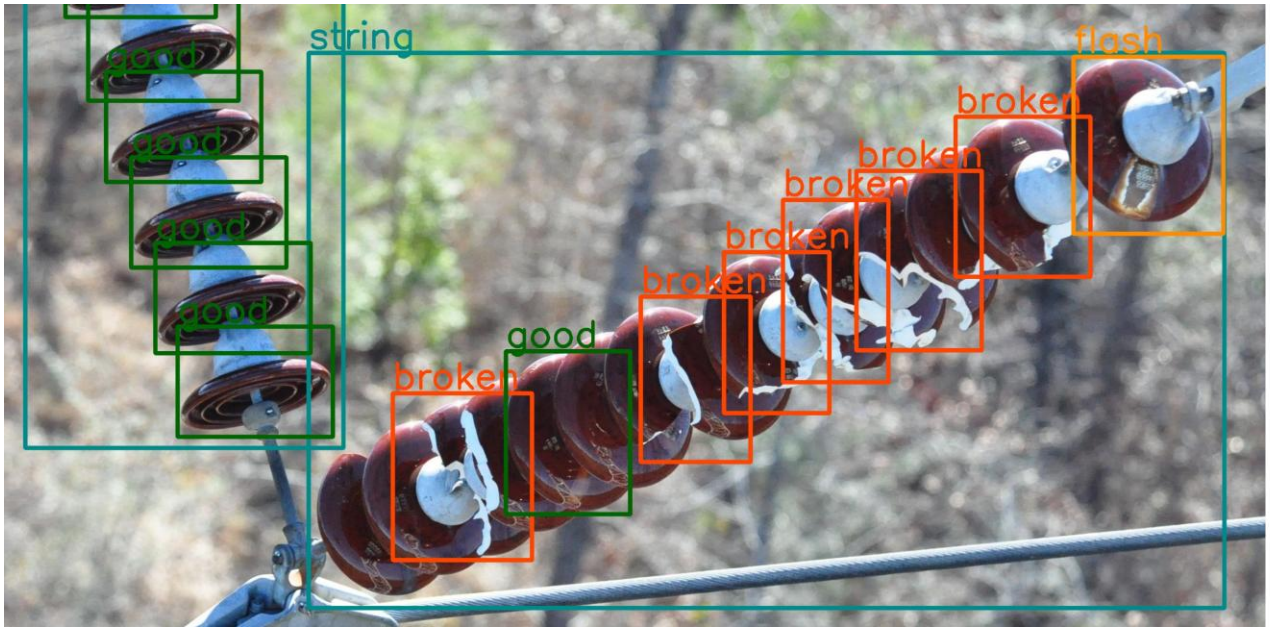
- Remove Mirrored Images



Dexter Lewis and Pratik Kulkarni, 'EPRI Insulator Defect Image Dataset'. IEEE DataPort, Aug. 11, 2021. doi: 10.21227/VKDW-X769

Curated Dataset

488 Images



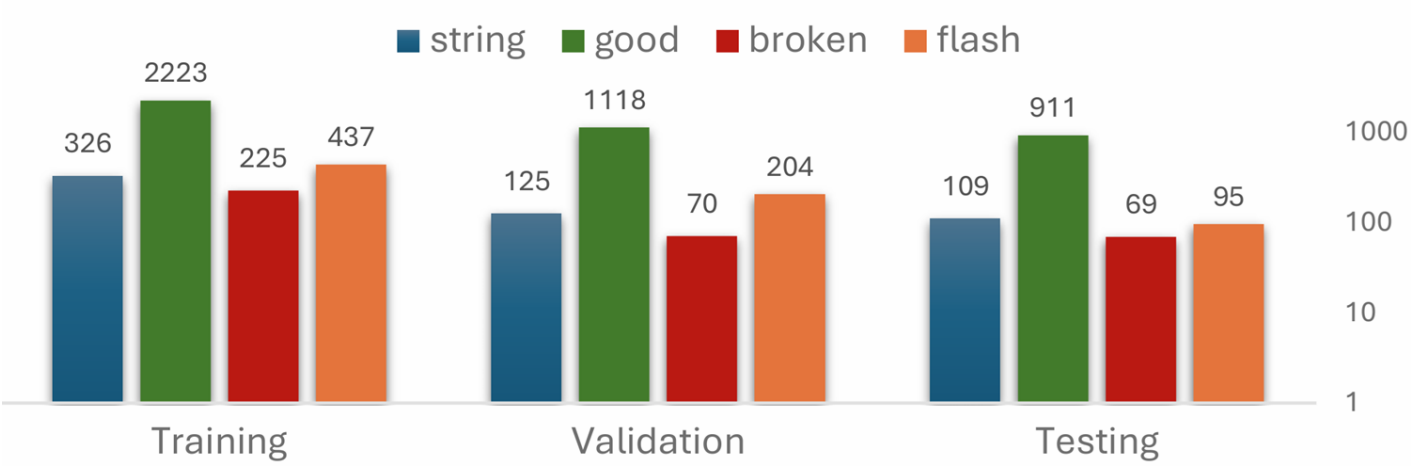


560 String **4252 Good** **364 Broken** **736 Flash**

https://universe.roboflow.com/academic-it/epri-insulators/analytics

Dataset Split

300 + 100 + 88 Images



Sub-Set	Training	Validation	Testing
Images	61.5 %	20.5 %	18.0 %
Objects	54.3 %	25.7 %	20.0 %

https://universe.roboflow.com/academic-it/epri-insulators/analytics

RESULTS

- Caption Tasks
- Region Tasks
- Segmentation
- Object Detection

Detailed Caption Results – Example 1



Model	Inference Results for Detailed Caption – Example 1
PT Florence	”A tall stack of chocolate and white candies on the ground.”, ”There is a tall stack of brown and white balls on the ground. The ground is brown and has green grass on it. There is a small white ball in the middle of the stack of balls.”
FT Florence	”goods in a line in front of a blurry background”, ”goods are goods. goods are browns are goods.goods have goods on them. goodgoods isgoods andgoods”
GPT4-o	”A vertical string of dark brown ceramic insulator discs with a glossy finish, separated by light gray cylindrical spacers. The insulator chain is part of a high-voltage power line, with a white, irregularly shaped f lash mark or residue on one of the discs, possibly indicating a discharge or damage. The background consists of a blurred ground with patches of green vegetation and brown soil.”

Detailed Caption Results – Example 2

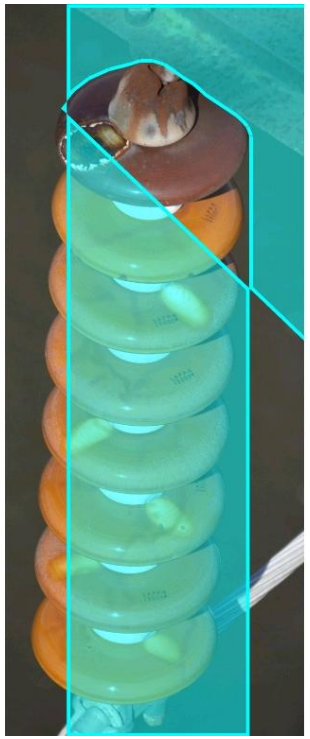


Model	Inference Results for Detailed Caption – Example 2
PT Florence	”There is a metal pole with a white cord attached to it. There are brown beads on the top of the pole. There is a brown piece of metal on top of them.”
FT Florence	”A string is hanging from a cord. The cord is white. There are six strings. The string is made of metal. The metal is gray. The lights are yellow. The flash is yellow.”
GPT4-o	”High-voltage ceramic insulator string with multiple brownish-orange disc insulators, connected to a transmission line. The insulators show visible signs of damage, including burn marks and erosion, especially on the top disc. Yellowish deposits are present on several discs, which may indicate contamination or material degradation. The assembly is secured to a metal crossarm, which has rust and corrosion, and the conductor is attached through a steel cable.”

Referring expression to segmentation – PT

text_input = "string insulators disc, good or defective"

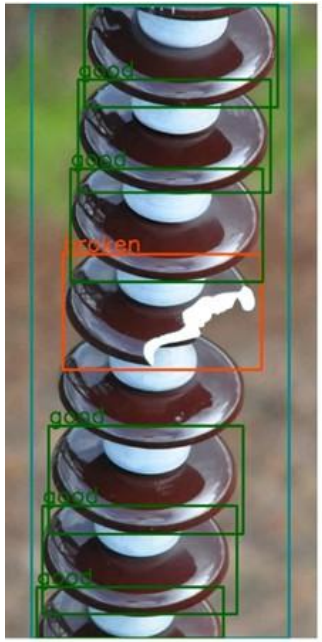


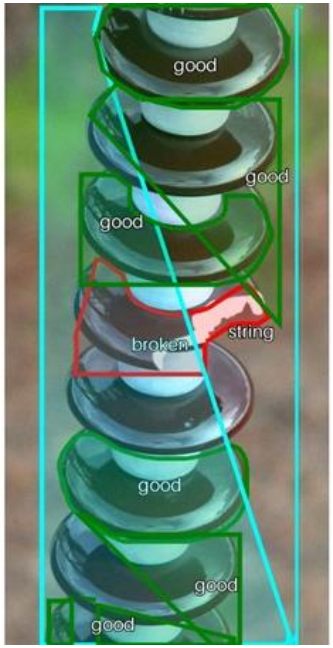



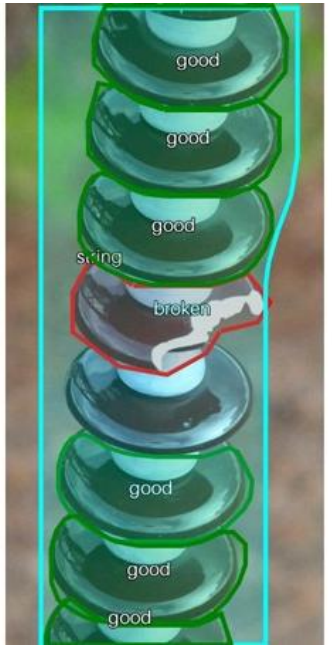
Regions to segmentation – Example 1

text_input = "class_id<loc_x1><loc_y1><loc_x2><loc_y2> (...)"





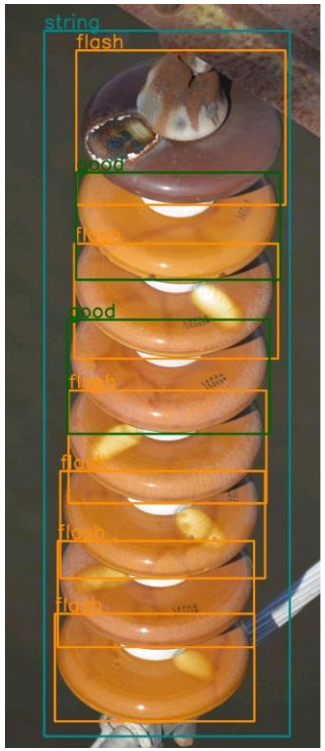


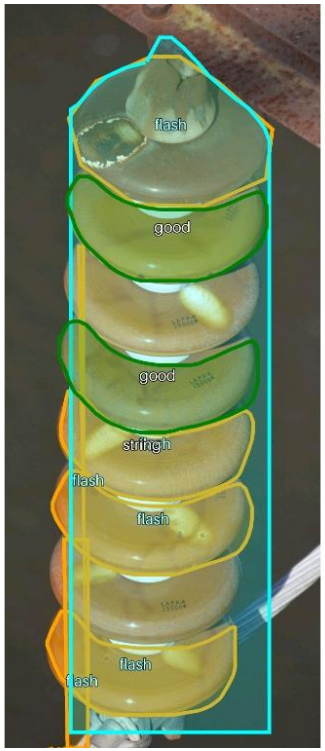


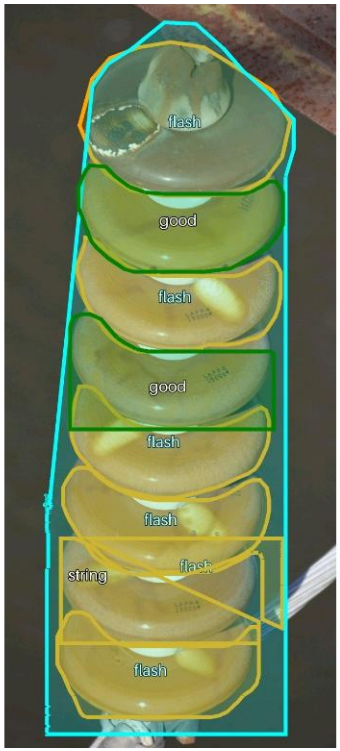
Labels **PT** **FT**

Regions to segmentation – Example 2



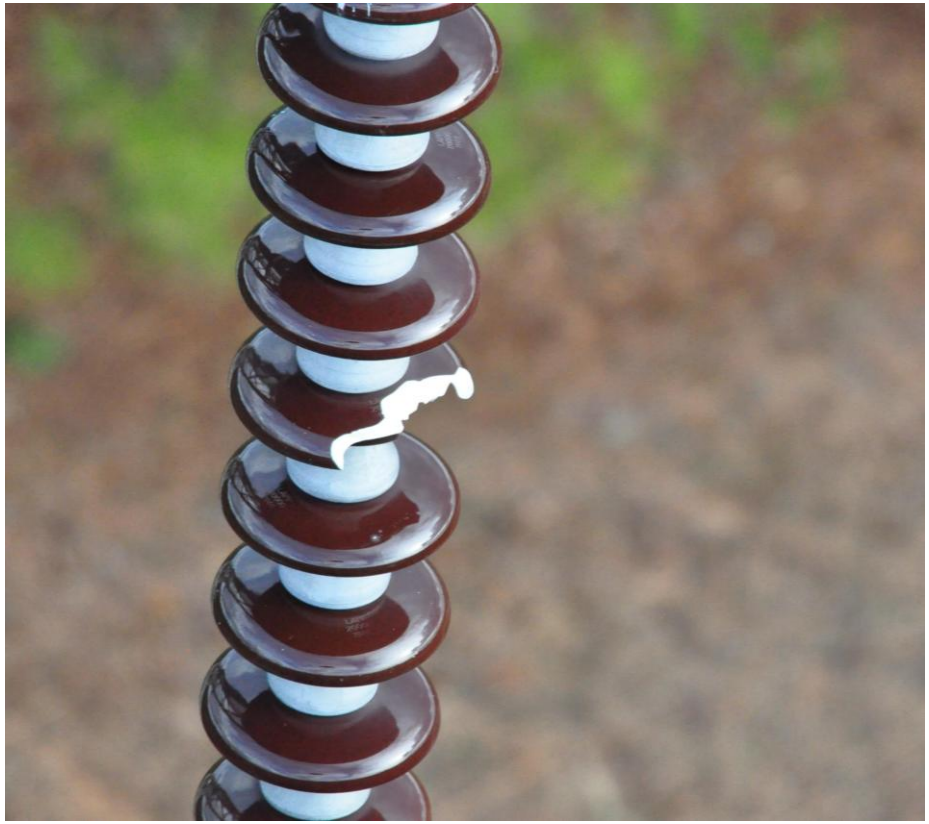


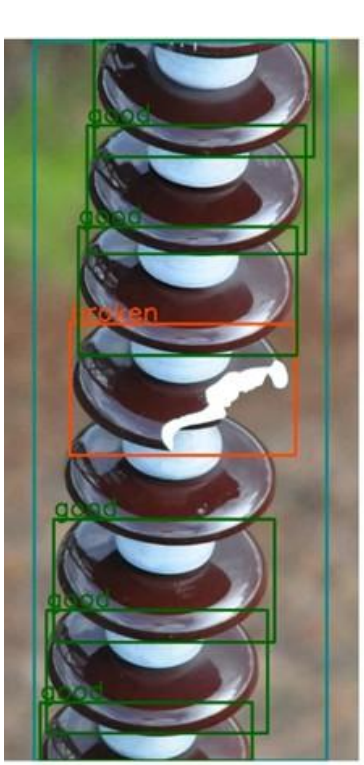


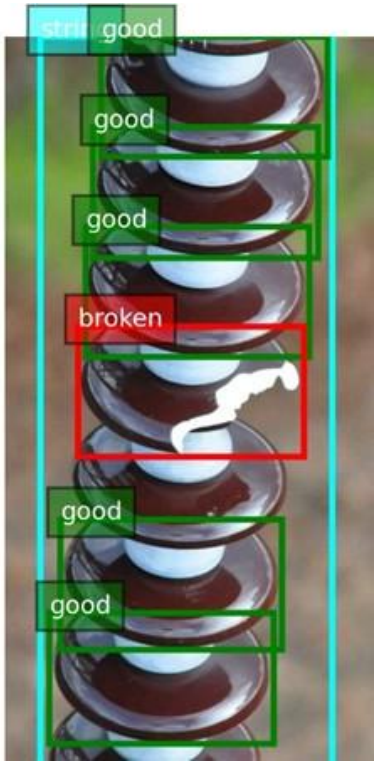


Labels	PT	FT

Object Detection – Example 1

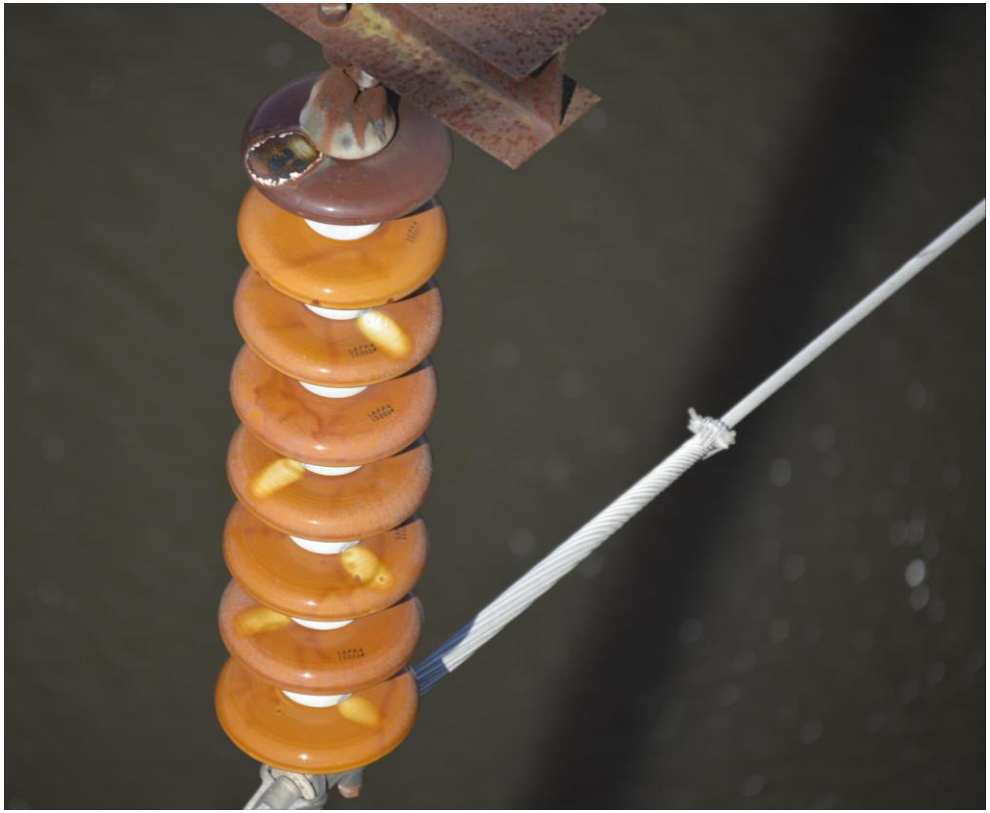


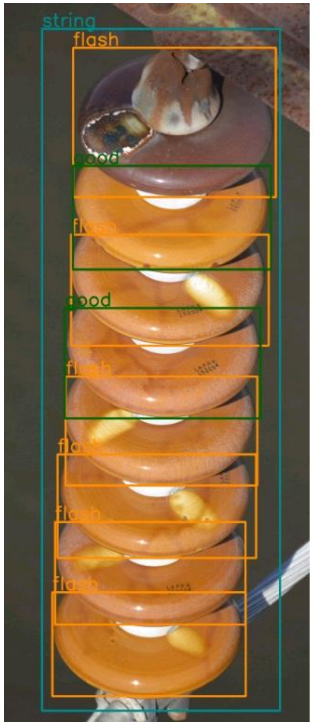




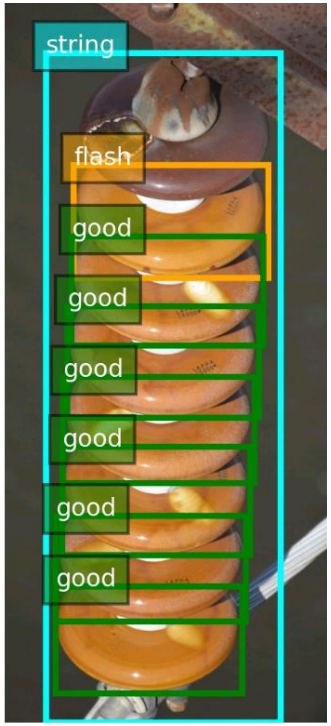
Labels FT

Object Detection – Example 2





Labels

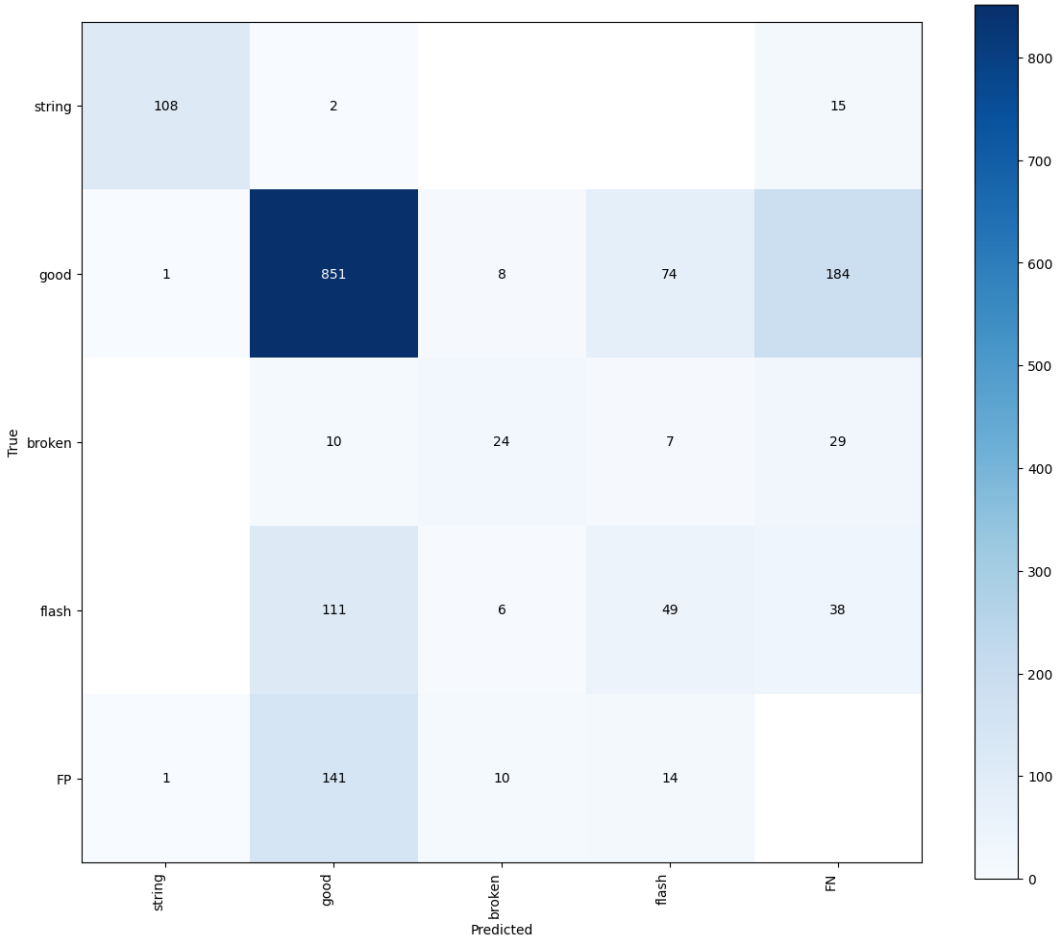


FT

Object Detection

Mean Average Precision

- mAP@50 = 0.536
- mAP@75 = 0.513
- mAP@50:95 = 0.461



MLLM Florence-2 VS CNN-based YOLO

Work	Model	# Params	mAP@50
Ours	Florence-2-base	0.23B	53.6%
[1]	YOLOv8s	11.2M	89.7%
[2]	YOLO11xl	56.9M	94.8%

[1] F. Lopes, P. Rocha and A. Coelho, "Towards Automated Visual Inspection of Electrical Grid Assets for the Smart Grid - An Application to HV Insulators," 2024 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Oslo, Norway, 2024, pp. 347-352, doi: 10.1109/SmartGridComm60555.2024.10738034.

[2] P. D. Rocha, L. A. S. Cruz, A. Coelho, and F. Lopes, ‘‘Automating electrical grid asset inspection using deep vision: Evaluating the impact of annotation detail level,’’ 2025

Conclusion

- Summary
- Challenges
- Limitations
- Contributions

SUMMARY

- MLLMs Potential
  - Large Architectures
  - Large Range of Tasks
- Challenges
  - PT on General Data
  - Lack Specific Datasets



Future Work

- More Experiments
  - Other Datasets
  - Other MLLMs
- Enrich Annotations
  - Specific Vocabulary
  - Spatial Definition



Q&A

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